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# A Spherical Rotation Coordinate System for the Description of Three-Dimensional Joint Rotations

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**Abstract—**Three-dimensional joint rotations in human movement analysis have been mainly described by Euler/Cardan angles. Due to sequence dependence, each combination of three Euler/Cardan angles defines a single pattern of joint rotation. When the rotation pattern is unknown, it needs to be considered using a particular sequence of Euler/Cardan angles to represent joint rotations. In this paper a spherical rotation coordinate system is developed for describing three-dimensional joint rotations using a method of rotation involving two steps: a long axis rotation and a pure axial rotation. Two angles of the classical spherical coordinate system—longitude and latitude—are used to describe long axis rotations in this newly proposed coordinate system. The spherical rotation coordinate system uses a radial rotation angle to describe pure axial rotation of a limb segment whereas the classical spherical coordinate system uses a radial displacement to describe motion of a point. An application of the spherical rotation coordinate system is given to define three-dimensional rotations of the glenohumeral joint. A mathematical proof shows that the long axis rotation and axial rotation are sequence independent. Two numerical examples are investigated which demonstrate that the spherical rotation angles can be uniquely determined in both forward and inverse kinematics without considering sequences of rotations. *© 2000 Biomedical Engineering Society.* [S0090-6964(00)00211-3]

**Keywords—**Biomechanics, Human movement, Three-dimensional rotation, Joint kinematics.

## INTRODUCTION

The development of modern technology now allows the recording of three-dimensional human movements. Analysis of the recorded information of motion has been advanced from two dimensions to three dimensions. One of the primary objectives of the analysis is to identify the pattern of joint or limb segment rotations from recorded information of motion. Since joint rotation is caused by motor controlled muscle activities, identification of real patterns of joint motion is critically important for understanding the mechanism of control in human movements and for understanding causes of musculoskeletal injuries. An important procedure in the analysis is to represent three-dimensional rotation angles of joints or limb segments. Three major techniques that are currently used for representing kinematic data are the Euler/Cardan angles,[1,5,14] the instantaneous screw axis method,[11,15] and a floating axis method.[2,7]

Sequence-dependent Euler/Cardan angles can produce different patterns of rotations depending upon the prespecified orders of rotation. When a set of recorded kinematic data is represented by two different sequences of Euler/Cardan angles (for example, a sequence of $\mathbf{R}_x \rightarrow \mathbf{R}_y \rightarrow \mathbf{R}_z$ rotation and a sequence of $\mathbf{R}_y \rightarrow \mathbf{R}_x \rightarrow \mathbf{R}_z$ rotation), the three rotation angles calculated by the two sequences may be different. Therefore, the results from Euler/Cardan angles need to be carefully interpreted.

The instantaneous screw axis method describes a translation along and a rotation about a screw axis at each instant of time. This method has been successfully used in engineering and mathematics because of uniqueness of the results and mathematical advantages. However, the screw angle or the rotation vector needs to be properly decomposed in an anatomical frame in clinical applications.[8,9]

Based on geometrical considerations, the joint coordinate system[2,7] with a floating axis has been used to determine the three rotation angles at the knee joint without considering the order of rotations. However, the option in choosing the two body fixed axes and a floating axis has been considered as imposing sequence effect under geometrical view.[15] The results from the joint coordinate system and a sequence of Cardan angles have been reported to be identical.[4,12] In addition, the geometrical characteristics of other joints, such as the shoulder joint, may not allow the floating axis to be properly located, which limits its application.

This problem of describing three-dimensional joint rotations may be solved by using three independent angles to describe three-dimensional joint rotations, in which one angle can change independently when the other two angles are fixed. In a recent study, Cheng *et al.*[3] proposed that a three-dimensional rotation of a limb seg-

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ment or joint can be decomposed into two sequence independent rotations, a long axis rotation and an axial rotation about the long axis. Therefore, a pure axial rotation angle of the limb segment between two recorded attitudes can be uniquely determined in a sense of two-step rotation.

The purpose of this paper is to further expand the two-step rotation method[3] into a spherical rotation coordinate system to describe the three-dimensional joint rotations with three independent rotation angles.

## THE SPHERICAL ROTATION COORDINATE SYSTEM

It is a common practice to describe the movement of a point in a three-dimensional space using three independent coordinates, such as *x, y, z* in a Cartesian coordinate system or longitude, latitude, radial displacement in a spherical coordinate system. When the radial displacement is a constant, the longitude and latitude in the spherical coordinate system describe an orbited movement of a point on the surface of a sphere or rotations of a point about a fixed position (the origin of the spherical coordinate system). Similar movement can be observed in the movement of human limb segments. Since the length of the long axis of a limb segment is a constant, motion of the distal end of the segment can be considered as moving on the surface of a sphere that is centered at the proximal end of the limb segment. The radius of the sphere is the length of the segment. Also the rotation of the long axis of the limb segment can be considered as a rotation about the proximal end (center of the sphere) of the limb segment. Therefore, in this paper, the longitude and latitude in the spherical coordinate system are used to describe the long axis rotation of the limb segment. Since the long axis is always in the radial direction, the axial rotation angle about the long axis is defined as the third angle and is called the radial rotation angle. This definition of axial rotation angle is identical to clinical descriptions.[10] The three angles (longitude $\alpha$, latitude $\beta$, radial rotation angle $\gamma$) form a new coordinate system (Fig. 1) for representing three-dimensional rotation of the limb segment or joint. There are two differences between the new coordinate system and the classical spherical coordinate system. One is that the radial rotation angle is used instead of radial displacement, and the other is that the new coordinate system is used to describe the three-dimensional rotations of a limb segment (rigid body) rather than the movement of a point. To differentiate this from the classical spherical coordinate system, the new coordinate system is called a *spherical rotation coordinate system*.

To determine the three rotation angles ($\alpha$, $\beta$, $\gamma$) in the spherical rotation coordinate system, two sets of Cartesian coordinate systems need to be properly defined in order that descriptions of rotation can be easily interpreted and understood by clinicians. The first coordinate system is a reference coordinate system. The second coordinate system is a body-fixed coordinate system that has one axis represents the long axis of the limb segment. As an example, an application of the spherical rotation coordinate system to the glenohumeral joint is given in the following section to show how to determine the three angles from recorded information of rotations.

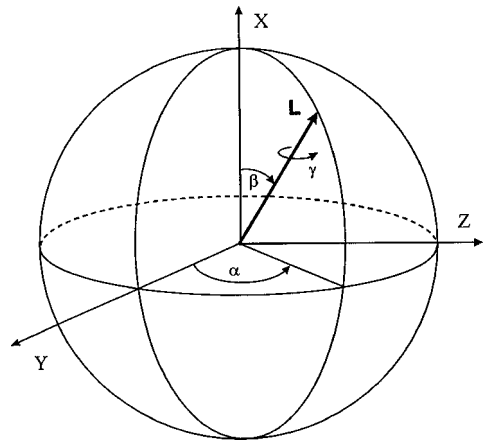


**FIGURE 1. A spherical rotation coordinate system includes three angles: longitude $\alpha$, latitude $\beta$, and radial rotation angle $\gamma$. L is a unit vector of the long axis of a limb segment. Longitude $\alpha$ and latitude $\beta$ are used to describe long axis rotation. Radial rotation angle $\gamma$ is used to describe a pure axial rotation about the long axis L of the limb segment. *XYZ* is a reference coordinate system.**

## APPLICATION TO THE GLENOHUMERAL JOINT

In this section the rotation of the right glenohumeral joint was considered as humeral rotation about the shoulder joint. The reference coordinate system *XYZ* (Fig. 2) was fixed on the trunk and referred to as the trunk coordinate system; the *X* axis directed downward, the *Y* axis directed forward, and the *Z* axis directed from left to right. The *XY* plane represented a sagittal plane, the *XZ* plane represented a frontal plane, and the *YZ* plane represented a horizontal plane. The second coordinate system *xyz* was fixed on the humerus (referred to as the humeral coordinate system) with the *x* axis representing the long axis of the humerus directed from the shoulder joint to the elbow joint, the *z* axis passing through the elbow joint directed laterally when the arm was at free hanging position with palm facing the body, and the *y* axis was perpendicular to the *x* and *z* axes by the right hand rule. For the convenience of illustration the *x, y, z* axes are not shown in Fig. 2.

It was assumed that the humeral coordinate system (*xyz*) was initially parallel to the trunk coordinate system

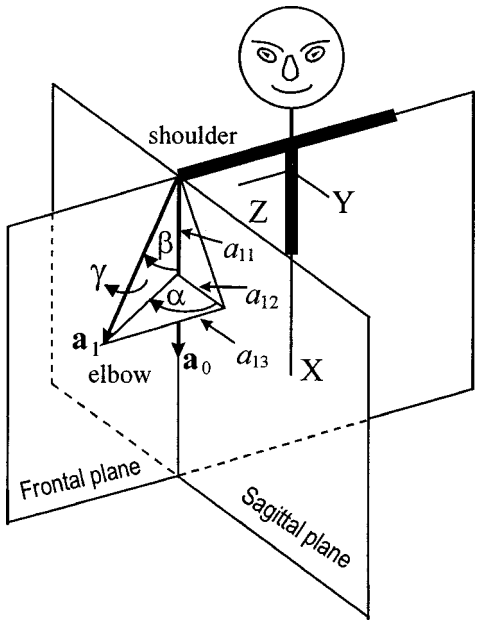


**FIGURE 2. An application of the spherical rotation coordinate system to the right glenohumeral joint. *XYZ* is the trunk coordinate system, defined such that *X* directs downward, *Y* directs forward, and *Z* directs from left to right. $\mathbf{a}_1$ is the unit vector of the *x* axis of the humeral coordinate system and it directs from the shoulder joint to the elbow joint. Angle $\beta$ represents elevation of the humerus, angle $\alpha$ defines a plane of the elevation, and angle $\gamma$ represents a pure axial rotation of the humerus. $\mathbf{a}_0$ is the unit vector of the long axis (*x* axis) of the humerus at initial position. $a_{11}$, $a_{12}$, $a_{13}$ are components of $\mathbf{a}_1$.**

(*XYZ*). This attitude of the humerus is referred to as neutral attitude and the long axis at the neutral attitude is denoted as $\mathbf{a}_0$ in Fig. 2. From the initial neutral position the long axis $\mathbf{a}_0$ elevated an angle $\beta$ to $\mathbf{a}_1$ in a plane defined by angle $\alpha$. During the elevation, the humerus axially rotated an angle $\gamma$ about the long axis. The elevation angle $\beta$ is measured from the vertical axis (*X* axis) of the trunk (reference) frame to the long axis $\mathbf{a}_1$ of the humerus. The angle $\alpha$ defines a plane of elevation, and is measured in the horizontal plane (*YZ* plane) from the *Y* axis of the trunk coordinate system to the projection of the long axis on the horizontal plane. The angle $\gamma$ is a pure axial rotation angle about the long axis of the humerus between the two attitudes and it can occur before the elevation when the long axis is at position $\mathbf{a}_0$ or occur after the elevation when the long axis is at position $\mathbf{a}_1$. The three angles, elevation angle $\beta$, angle $\alpha$ of the elevation plane, and the axial rotation angle $\gamma$, are identical to the latitude, longitude, and radial rotation angle of the spherical rotation coordinate system.

It is assumed that the earlier pattern of rotation has been recorded by a motion analysis system. Therefore the direction cosine matrix of the humeral coordinate system *xyz* with respect to the trunk coordinate system *XYZ* can be found as

$$\mathbf{D}_{h-t}(t)=\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}=\begin{bmatrix} \mathbf{a}_1\cdot\mathbf{e}_1 & \mathbf{a}_1\cdot\mathbf{e}_2 & \mathbf{a}_1\cdot\mathbf{e}_3 \\ \mathbf{a}_2\cdot\mathbf{e}_1 & \mathbf{a}_2\cdot\mathbf{e}_2 & \mathbf{a}_2\cdot\mathbf{e}_3 \\ \mathbf{a}_3\cdot\mathbf{e}_1 & \mathbf{a}_3\cdot\mathbf{e}_2 & \mathbf{a}_3\cdot\mathbf{e}_3 \end{bmatrix}, \tag{1}$$

$$\text{where}\quad \mathbf{a}_1=\begin{bmatrix} a_{11} \\ a_{12} \\ a_{13} \end{bmatrix},\quad \mathbf{a}_2=\begin{bmatrix} a_{21} \\ a_{22} \\ a_{23} \end{bmatrix},\quad \mathbf{a}_3=\begin{bmatrix} a_{31} \\ a_{32} \\ a_{33} \end{bmatrix} \tag{2}$$

are the three unit vectors of the *x*, *y*, *z* axes with respect to the trunk coordinate system, respectively. $\mathbf{e}_1,\mathbf{e}_2,\mathbf{e}_3$ are the three unit vectors of the *X*, *Y*, *Z* axes of the trunk frame, respectively. $a_{ij}$ is the *j*th component of the *i*th segment-fixed unit vector. Angles $\alpha$ and $\beta$ can be calculated from components $a_{11},a_{12},a_{13}$ of the measured unit vector $\mathbf{a}_1$ of the long axis as

$$\alpha=\arccos\left[\frac{a_{12}}{\sqrt{(a_{12})^2+(a_{13})^2}}\right] \qquad 0\leqslant\alpha\leqslant\pi, \tag{3}$$

$$\beta=\arccos(a_{11}) \qquad 0\leqslant\beta\leqslant\pi. \tag{4}$$

Considering the sign of $a_{13}$, a component of the unit vector $\mathbf{a}_1$ on the transverse axis (*Z* axis) of the trunk frame, and the long axis position in the trunk frame the sign of rotation angle $\alpha$ in Eq. (3) is defined as

$$\alpha>0 \qquad \text{when}\ \ a_{13}>0, \tag{5}$$

$$\alpha<0 \qquad \text{when}\ \ a_{13}<0. \tag{6}$$

The elevation angle $\beta$ ranges from 0° to 180° and the angle $\alpha$ of elevation plane ranges from −180° to 180°. When the long axis is at initial position, the elevation angle $\beta$ is zero. This can be calculated from Eq. (4), since $a_{11}=1$. However, when the elevation angle is zero, the components $a_{12}$ and $a_{13}$ are zero too. The angle $\alpha$ can not be directly computed from Eq. (3). In fact, when the elevation angle is zero, the value of angle $\alpha$ (defining a plane of elevation) does not have a practical meaning. Therefore, for convenience of illustration angle $\alpha$ can be assigned to zero when the long axis is at the neutral position and at the opposite of the neutral position.

Out of the neutral position, the elevation angle $\beta$ can change in an elevation plane defined by angle $\alpha$, also the angle $\alpha$ can change independently when the elevation angle $\beta$ is fixed. Therefore, the two angles describing the long axis rotations are independent.

**TABLE 1. Recommended description for the long axis positions of the humerus in the trunk frame.**

| Long-axis position | $a_{12}>0$ in front of the trunk | $a_{12}=0$ in frontal plane | $a_{12}<0$ at back of the trunk |
|---|---|---|---|
| $a_{13}>0$ at right side of the shoulder joint | $\alpha>0$: lateral rotation $\beta$: forward elevation | $\alpha=90°$ lateral rotation $\beta$: abduction | $\alpha>0$: lateral rotation $\beta$: backward elevation |
| $a_{13}=0$ in sagittal plane | $\alpha=0°$ $\beta$: flexion | $\alpha=0°$ $\beta=0°$ or 180° | $\alpha=0°$ $\beta$: extension |
| $a_{13}<0$ at left side of the shoulder joint | $\alpha<0$: medial rotation $\beta$: forward elevation | $\alpha=90°$ medial rotation $\beta$: adduction | $\alpha<0$: medial rotation $\beta$: backward elevation |

Alternatively, the positional angles $\alpha$, $\beta$ can be illustrated with words as recommended in Table 1 according to the long axis position in different coordinate phases of the trunk frame, and according to the combination of the signs of $a_{12}$ and $a_{13}$. For instance, when the humerus is pointing in front of the trunk ($a_{12}>0$) and to the right side of the shoulder joint ($a_{13}>0$), the long axis position can be described by a lateral rotation angle $\alpha$ and a forward elevation angle $\beta$. When the humerus is pointing in front of the trunk ($a_{12}>0$) and to the left side of the shoulder joint ($a_{13}<0$), its long axis position can be described as a medial rotation angle $\alpha$ and a forward elevation angle $\beta$. When the humerus is pointing behind the trunk, a backward elevation is used to differentiate from forward elevation in the front of the trunk.

In a special case, the spherical rotation coordinate system describes two-dimensional joint rotations used by clinicians. When the angle of the elevation plane is zero ($\alpha=0$, $a_{13}=0$), elevation angle $\beta$ describes a flexion/extension since it occurs in the sagittal plane. When the angle of elevation plane is 90° ($\alpha=90°$, $a_{12}=0$), the elevation angle $\beta$ is equivalent to abduction/adduction in the frontal plane (Table I).

When the long axis directs upward in the opposite direction of the $X$ axis, the elevation angle can be determined by Eq. (4) since $a_{11}=-1$, and the angle $\alpha$ can be assigned to a value of zero as stated earlier.

The axial rotation angle $\gamma$ can be determined using a method of two-step rotation,[3] which is summarized as follows. The rotation of the long axis from initial position $\mathbf{a}_0$ to final position $\mathbf{a}_1$ can be considered as a rotation about an axis with a unit vector $\mathbf{n}$ that is perpendicular to the elevation plane $\Pi$ (Fig. 3) where the long axis rotation occurs. The unit vector $\mathbf{n}$ can be determined by vector product of $\mathbf{a}_0$ and $\mathbf{a}_1$ as

$$\mathbf{n}=\mathbf{a}_0\times\mathbf{a}_1/|\mathbf{a}_0\times\mathbf{a}_1|. \tag{7}$$

The rotation matrix for the long axis rotation is[3]

$$\mathbf{R}_l(\beta,\mathbf{n})=\cos\beta\, I+(1-\cos\beta)[n][n]^T+\sin\beta[\tilde{n}]. \tag{8}$$

In which $\mathbf{I}$ is a $3\times3$ unit matrix, $[\tilde{n}]$ is a skew matrix composed of the components of unit vector $\mathbf{n}$ and is written as

$$[\tilde{n}]=\begin{vmatrix} 0 & -n_3 & n_2 \\ n_3 & 0 & -n_1 \\ -n_2 & n_1 & 0 \end{vmatrix}. \tag{9}$$

The orientation of the $y$ axis of the body-fixed coordinate system following the long axis rotation can be found using the rotation matrix (Eq. 8) as

$$\mathbf{v}'=\mathbf{R}_l(\beta,\mathbf{n})\mathbf{v}_0, \tag{10}$$

where $\mathbf{v}_0=\mathbf{e}_2$ is the unit vector of the $y$ axis at initial attitude before the long axis rotation, $\mathbf{v}'$ is the unit vector of the $y$ axis after the long rotation. Since the unit vector $\mathbf{v}_1=\mathbf{a}_2$ [Eq. (2)] of the $y$ axis at final attitude has been recorded, the pure axial rotation angle can be calculated as

$$\gamma=\arccos(\mathbf{v}'\cdot\mathbf{v}_1/|\mathbf{v}'||\mathbf{v}_1|). \tag{11}$$

The sign of angle $\gamma$ can be determined as

$$\gamma>0 \qquad \text{if } 0<\mathbf{a}'\cdot\mathbf{a}_1\leqslant1, \tag{12}$$

$$\gamma<0 \qquad \text{if } -1\leqslant\mathbf{a}'\cdot\mathbf{a}_1<0, \tag{13}$$

where $\mathbf{a}'=\mathbf{v}'\times\mathbf{v}_1/|\mathbf{v}'\times\mathbf{v}_1|$.

## ROTATION MATRIX

Given a convention of rotation without considering translations, there exists a corresponding rotation or

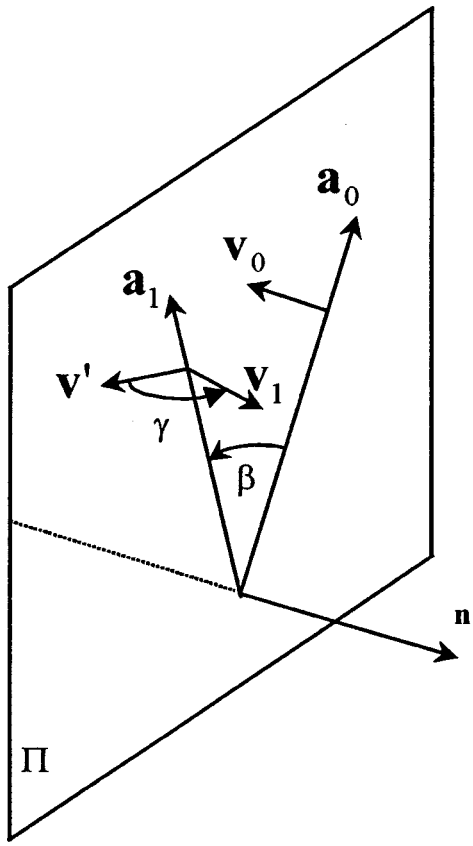


**FIGURE 3. Unit vectors $\mathbf{a}_0$ and $\mathbf{a}_1$ present initial and final positions of long axis of a limb segment in a reference coordinate system, respectively. Unit vector n is the rotation axis of the long axis rotation from $\mathbf{a}_0$ to $\mathbf{a}_1$, and $\beta$ is the long axis rotation angle. $\mathbf{v}_0$, $\mathbf{v}'$, and $\mathbf{v}_1$ are the unit vectors of an axis (*y* axis) of body-fixed coordinate system. The *y* axis moves from initial position $\mathbf{v}_0$ to position $\mathbf{v}'$ with the long axis rotation from $\mathbf{a}_0$ to $\mathbf{a}_1$, then it rotates axially an angle $\gamma$ about the long axis $\mathbf{a}_1$ from $\mathbf{v}'$ to $\mathbf{v}_1$. If the unit vector $\mathbf{v}'$ is known, then the axial rotation angle $\gamma$ can be determined from the scalar product of the unit vectors $\mathbf{v}'$ and $\mathbf{v}_1$.**

transformation matrix that is usually used to relate the location of a point in the body-fixed coordinate system to its position in the reference coordinate system.[6] The rotation matrix can also relate the orientations of a body-fixed vector or body-fixed frame in the reference frame before and after rotation.[13] For the $x-y-z$ sequence of the Euler/Cardan rotation (first rotate an angle $\theta$ about the $x$ axis, then rotate an angle $\varphi$ about the $y$ axis, and finally rotate an angle $\psi$ about the $z$ axis) the integrated rotation matrix $\mathbf{R}(\theta,\varphi,\psi)$ has the following form:

$$\mathbf{R}(\theta,\varphi,\psi)=\mathbf{R}_z(\psi)\mathbf{R}_y(\varphi)\mathbf{R}_x(\theta), \tag{14}$$

where $\mathbf{R}_x(\theta)$, $\mathbf{R}_y(\varphi)$, $\mathbf{R}_z(\psi)$, are the rotation matrices about the $x$, $y$, and $z$ axes of the body-fixed coordinate system, respectively. The integrated rotation matrix $\mathbf{R}(\theta,\varphi,\psi)$ is formed by multiplication of the three rotation matrices $\mathbf{R}_x(\theta)$, $\mathbf{R}_y(\varphi)$, $\mathbf{R}_z(\psi)$ in inverse order of the three rotations. If the order of the rotation is first about the $z$ axis, then about the $y$ axis, and finally about the $x$ axis the integrated rotation matrix will be $\mathbf{R}(\theta,\varphi,\psi)=\mathbf{R}_x(\theta)\mathbf{R}_y(\varphi)\mathbf{R}_z(\psi)$. Since, generally, the product of two matrices is not commutative, changing the order of occurrence of the three rotations will result in a different integrated rotation matrix that represents a new sequence of rotation. This characteristic is called *sequence dependence*. Due to sequence dependence, given a set of three rotation angles about axes of a body-fixed coordinate system a joint can rotate from one attitude to two different attitudes by two different sequences of rotations. But if two sequences of rotation are equivalent, or the rotations are *sequence independent*, then the two integrated rotation matrices must be the same. Therefore, for a given set of rotation angles a joint can rotate (or transfer) from one attitude to another by different sequences of rotations.

The integrated rotation matrix of a rotation sequence involving less than or more than three primary rotations can also be derived in the same way as illustrated earlier. In this paper, the three-dimensional rotation of a limb segment or joint is considered to be composed of two primary rotations—a long axis rotation and an axial rotation. There are two possible combinations for the two rotation sequences. The first is an axial rotation followed by a long axis rotation. In this sequence, the axial rotation of an angle $\gamma$ is about the long axis at initial position $\mathbf{a}_0$, and the long axis rotation of angle $\beta$ is from initial position $\mathbf{a}_0$ to position $\mathbf{a}_1$ about an rotation axis with an unit vector $\mathbf{n}$ in the reference coordinate system $XYZ$ (Fig. 4). The second rotation sequence consists of a long axis rotation followed by an axial rotation. The long axis rotation in this case is the same as in the first, but the axial rotation of an angle $\gamma$ is about the long axis at displaced position $\mathbf{a}_1$. In the follow paragraphs we will derive an integrated rotation matrix for each of the two rotation sequences and show that the two integrated rotation matrixes are the same. Therefore, the sequence independence of the long axis rotation and the axial rotation can be proved and a unique integrated rotation matrix can be obtained.

The integrated rotation matrix of each rotation sequence can be obtained by multiplication of two rotation matrices: one rotation matrix describing the long axis rotation and another rotation matrix describing the axial rotation. We first investigate the long axis rotation to obtain its rotation matrix. The long axis rotation (Fig. 4) from initial position $\mathbf{a}_0$ to position $\mathbf{a}_1$ by an angle $\beta$ about axis $\mathbf{n}$ can be described by a rotation matrix $\mathbf{R}_l(\beta,\mathbf{n})$ as

$$\mathbf{a}_1=\mathbf{R}_l(\beta,\mathbf{n})\mathbf{a}_0, \tag{15}$$

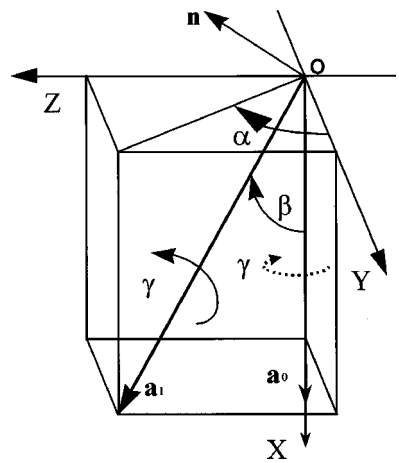


**FIGURE 4. The elevation of the long axis is considered as a rotation of the long axis from $\mathbf{a}_0$ to $\mathbf{a}_1$ about a rotation axis presented by unit vector n that is perpendicular to the plane formed by initial attitude $\mathbf{a}_0$ and final attitude $\mathbf{a}_1$ of the long axis, and is defined as $\mathbf{n}=\mathbf{a}_0\times\mathbf{a}_1/|\mathbf{a}_0\times\mathbf{a}_1|$. Axial rotation angle $\gamma$ can be about $\mathbf{a}_0$ before the long axis rotation or about $\mathbf{a}_1$ after the long axis rotation.**

where

$$\mathbf{a}_0=\mathbf{e}_1 \tag{16}$$

can be written as

$$\mathbf{a}_0=[a_0]=[e_1]=\begin{bmatrix}1\\0\\0\end{bmatrix}. \tag{17}$$

$\mathbf{e}_1$ is the unit vector of the $X$ axis. The matrix form of unit vector $\mathbf{a}_1$ can be written in terms of three components $a_{11}$, $a_{12}$, $a_{13}$ in the reference coordinate system or in terms of two angles $\alpha$, $\beta$ in the spherical rotation coordinate system as

$$\mathbf{a}_1=[a_1]=\begin{bmatrix}a_{11}\\a_{12}\\a_{13}\end{bmatrix}=\begin{bmatrix}\cos\beta\\\sin\beta\cos\alpha\\\sin\beta\sin\alpha\end{bmatrix}. \tag{18}$$

The unit vector $\mathbf{n}$ of the rotation axis of the long axis rotation is perpendicular to the plane formed by the $X$ axis and vector $\mathbf{a}_1$ and is defined as

$$\mathbf{n}=\mathbf{e}_1\times\mathbf{a}_1/|\mathbf{e}_1\times\mathbf{a}_1|. \tag{19}$$

Using Eqs. (17) and (18), the unit vector $\mathbf{n}$ in Eq. (19) can be expressed in matrix form as

$$\mathbf{n}=[n]=\begin{bmatrix}n_1\\n_2\\n_3\end{bmatrix}=\begin{bmatrix}0\\-\sin\alpha\\\cos\alpha\end{bmatrix}, \tag{20}$$

where $n_1$, $n_2$, $n_3$ are the three components of vector $\mathbf{n}$ in the reference coordinate system.

Two parameters $\beta$ and $\mathbf{n}$ in the rotation matrix $\mathbf{R}_l(\beta,\mathbf{n})$ indicate that rotation angle $\beta$ is about rotation axis $\mathbf{n}$, the subscript $l$ is used to indicate that the rotation is a long axis rotation. The rotation matrix $\mathbf{R}_l(\beta,\mathbf{n})$ has the general form of Eq. (8). Substituting the unit vector $\mathbf{n}$ of the rotation axis in Eq. (20) into Eq. (8), the rotation matrix of the long axis rotation can be expressed in terms of two spherical rotation angles $\alpha$ and $\beta$ as

$$\mathbf{R}_l(\beta,\mathbf{n})=\mathbf{R}_l(\alpha,\beta)=\begin{vmatrix}\cos\beta & -\sin\beta\cos\alpha & -\sin\beta\sin\alpha\\ \sin\beta\cos\alpha & \cos\beta+(1-\cos\beta)\sin^2\alpha & -(1-\cos\beta)\sin\alpha\cos\alpha\\ \sin\beta\sin\alpha & -(1-\cos\beta)\sin\alpha\cos\alpha & \cos\beta+(1-\cos\beta)\cos^2\alpha\end{vmatrix}. \tag{21}$$

Now we investigate the axial rotation of the limb segment to obtain its rotation matrix. Since the rotation matrix $\mathbf{R}_l(\beta,\mathbf{n})$ describes a rotation angle $\beta$ about an axis $\mathbf{n}$ that is defined in a three-dimensional reference frame, the rotation matrix in Eq. (8) can be used to describe rotations about any other axes in the reference frame. Therefore, we can use the rotation matrix $\mathbf{R}_l(\beta,\mathbf{n})$ to describe axial rotations by substitute a unit vector of the long axis in place of $\mathbf{n}$ and an axial rotation angle $\gamma$ in place of $\beta$. In the first sequence the axial rotation occurs before the long axis rotation and is about axis $\mathbf{a}_0$; therefore, rotation matrix of the axial rotation can be obtained by substituting $\mathbf{n}$ with $\mathbf{a}_0$ and $\beta$ with axial rotation angle $\gamma$:

$$\begin{aligned}\mathbf{R}_a(\gamma,\mathbf{a}_0)&=\mathbf{R}_a(\gamma,\mathbf{e}_1)\\&=\cos\gamma\mathbf{I}+(1-\cos\gamma)[a_0][a_0]^T+\sin\gamma[\tilde{a}_0],\end{aligned} \tag{22}$$

where subscript $a$ in $\mathbf{R}_a(\gamma,\mathbf{a}_0)$ indicates that the rotation matrix is for an axial rotation. Now, the integrated rotation matrix for the first sequence rotation can be constructed by multiplying the two rotation matrices $\mathbf{R}_l(\alpha,\beta)$ and $\mathbf{R}_a(\gamma,\mathbf{a}_0)$ in reverse order of the rotations as

$$\mathbf{R}_1(\alpha,\beta,\gamma)=\mathbf{R}_l(\alpha,\beta)\mathbf{R}_a(\gamma,\mathbf{a}_0), \tag{23}$$

where subscript 1 in $\mathbf{R}_1(\alpha,\beta,\gamma)$ indicates that the rotation matrix is for the first sequence of rotation—axial rotation followed by long axis rotation.

In the second sequence of rotation, the long axis rotation is the same as that in the first sequence: the segment moves from position $\mathbf{a}_0$ to position $\mathbf{a}_1$. However, the axial rotation is now about $\mathbf{a}_1$. Therefore, the rotation matrix for the axial rotation about $\mathbf{a}_1$ is

$$\mathbf{R}_a(\gamma,\mathbf{a}_1)=\cos\gamma\mathbf{I}+(1-\cos\gamma)[a_1][a_1]^T+\sin\gamma[\tilde{a}_1]. \tag{24}$$

In this rotation matrix the rotation axis has been changed to $\mathbf{a}_1$. The integrated rotation matrix for the second sequence of rotation becomes

$$\mathbf{R}_2(\alpha,\beta,\gamma)=\mathbf{R}_a(\gamma,\mathbf{a}_1)\mathbf{R}_l(\alpha,\beta), \tag{25}$$

where subscript 2 in $\mathbf{R}_2(\alpha,\beta,\gamma)$ denotes the second sequence of rotation. The rotation matrices $\mathbf{R}_1(\alpha,\beta,\gamma)$ and $\mathbf{R}_2(\alpha,\beta,\gamma)$ describe two different sequences of rotation, respectively. If the two sequences of rotation are equivalent, indicating that the long axis rotation and pure axial rotation are sequence independent, then the two rotation matrices must be the same. This result is obtained by using Eqs. (17)–(25):

$$\begin{aligned}\mathbf{R}(\alpha,\beta,\gamma)&=\mathbf{R}_l(\alpha,\beta)\mathbf{R}_a(\gamma,\mathbf{a}_0)\\&=\mathbf{R}_a(\gamma,\mathbf{a}_1)\mathbf{R}_l(\alpha,\beta)\\&=\begin{vmatrix} C\beta & -S\beta C\alpha C\gamma-S\beta S\alpha S\gamma & S\beta C\alpha S\gamma-S\beta S\alpha C\gamma\\ S\beta C\alpha & C\beta C\gamma+(1-C\beta)(S^2\alpha C\gamma-S\alpha C\alpha S\gamma) & -C\beta S\gamma-(1-C\beta)(S^2\alpha S\gamma+S\alpha C\alpha C\gamma)\\ S\beta S\alpha & C\beta S\gamma+(1-C\beta)(C^2\alpha S\gamma-S\alpha C\alpha C\gamma) & C\beta C\gamma+(1-C\beta)(C^2\alpha C\gamma+S\alpha C\alpha S\gamma)\end{vmatrix},\end{aligned} \tag{26}$$

where $S=\text{sine}$, $C=\text{cosine}$, and $\mathbf{R}(\alpha,\beta,\gamma)$ is the integrated rotation matrix. Therefore, it has been proven mathematically that the long axis rotation and the axial rotation are sequence independent. This result also demonstrates the independence of the three spherical rotation angles (longitude, latitude, and radial rotation angle), since the two long axis rotation angles (longitude and latitude) are independent. As shown earlier, the three angles $\alpha$, $\beta$, $\gamma$ span the following ranges:

$$-\pi\leqslant\alpha\leqslant\pi,\quad 0\leqslant\beta\leqslant\pi,\quad -\pi\leqslant\gamma\leqslant\pi. \tag{27}$$

Knowing a position of a vector $\mathbf{r}$ in the reference coordinate system, and three angles $\alpha$, $\beta$, $\gamma$ of the spherical rotation coordinate system, its position $\mathbf{r}'$ after the rotation can be found using the integrated rotation matrix in Eq. (26) as

$$\mathbf{r}'=\mathbf{R}(\alpha,\beta,\gamma)\mathbf{r}. \tag{28}$$

## NUMERICAL EXAMPLES

In this section, two simple different patterns of three-dimensional right humeral rotations are investigated to show that the spherical rotation coordinate system can be used in both inverse and forward kinematics to describe different patterns of joint rotations. The reference coordinate system and humeral coordinate system were previously defined (Figs. 2 and 4). Initially, the two coordinate systems were aligned with the $x$ axis directed vertically downward, $y$ axis forward, and $z$ axis from left to right. We use $\mathbf{u}^0,\mathbf{v}^0,\mathbf{w}^0$ to represent initial positions of the unit vectors of the $x$, $y$, $z$ axes in the reference frame, respectively. Therefore,

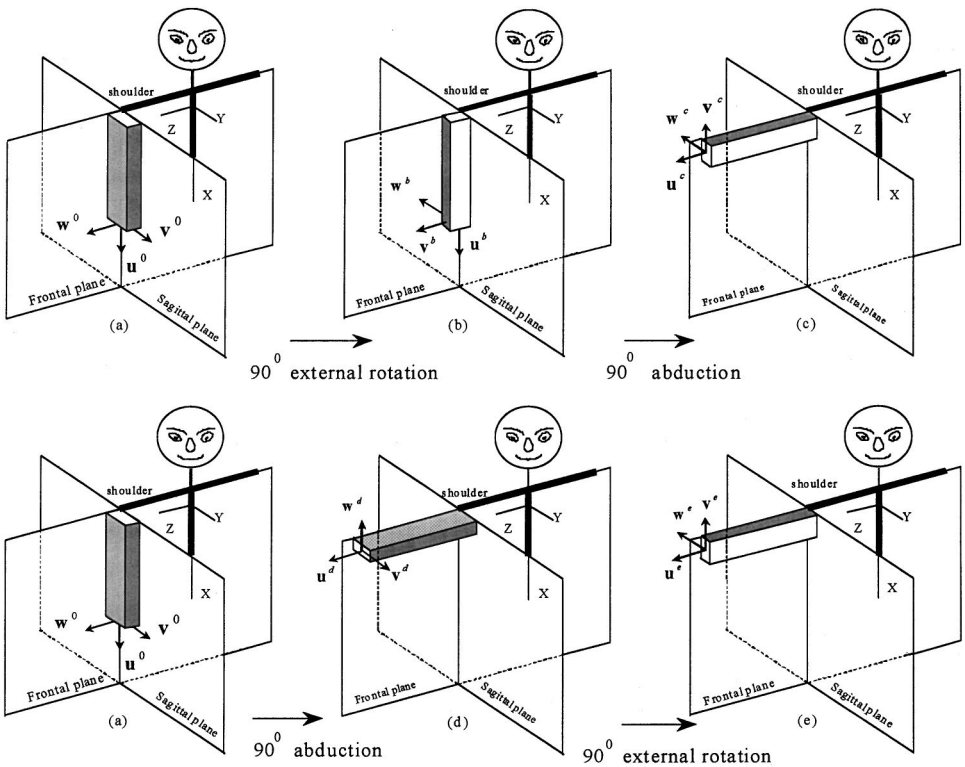


**FIGURE 5. A right humeral rotation involving 90° abduction and 90° external rotation can be performed in two sequences. In sequence 1, the first rotation is 90° external rotation from attitude $(\mathbf{u}^0,\mathbf{v}^0,\mathbf{w}^0)$ in (a) to attitude $(\mathbf{u}^b,\mathbf{v}^b,\mathbf{w}^b)$ in (b), then followed by 90° abduction to attitude $(\mathbf{u}^c,\mathbf{v}^c,\mathbf{w}^c)$ in (c). In sequence 2, the first rotation is 90° abduction from attitude $(\mathbf{u}^0,\mathbf{v}^0,\mathbf{w}^0)$ in (a) to attitude $(\mathbf{u}^d,\mathbf{v}^d,\mathbf{w}^d)$ in (d) followed by 90° external rotation to attitude $(\mathbf{u}^e,\mathbf{v}^e,\mathbf{w}^e)$ in (e). As shown in the figure the two sequences are equivalent, since final attitude $(\mathbf{u}^c,\mathbf{v}^c,\mathbf{w}^c)$ of sequence 1 equals to final attitude $(\mathbf{u}^e,\mathbf{v}^e,\mathbf{w}^e)$ of sequence 2. *XYZ* is a reference coordinate system fixed on the trunk.**

$$\mathbf{u}^0=\begin{bmatrix} u_1^0 \\ u_2^0 \\ u_3^0 \end{bmatrix}=\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix},\quad \mathbf{v}^0=\begin{bmatrix} v_1^0 \\ v_2^0 \\ v_3^0 \end{bmatrix}=\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix},\quad \mathbf{w}^0=\begin{bmatrix} w_1^0 \\ w_2^0 \\ w_3^0 \end{bmatrix}=\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}. \tag{29}$$

*Example 1*

In this example, a pattern of humeral rotation involves a 90° abduction and a 90° external rotation. In the spherical rotation coordinate system this pattern of rotation is described as a 90° elevation in the coronal plane, and a 90° axial rotation. The three spherical rotation angles are $\alpha=90°$, $\beta=90°$, and $\gamma=90°$. There are two possible sequences for a combination of the abduction and axial rotation; the first is 90° external rotation followed by 90° abduction; and the second is 90° abduction followed by 90° external rotation. Graphical representation of the pattern of rotation in Fig. 5 shows that final attitudes $(\mathbf{u}^c,\mathbf{v}^c,\mathbf{w}^c)$ and $(\mathbf{u}^e,\mathbf{v}^e,\mathbf{w}^e)$ of the two possible sequences of rotation are the same, which demonstrates the sequence independence of the long axis rotation and the axial rotation. As shown in Fig. 5, the final attitude $(\mathbf{u}^1,\mathbf{v}^1,\mathbf{w}^1)$ can be expressed in the reference coordinate system as

$$\mathbf{u}^1=\mathbf{u}^c=\mathbf{u}^e=\begin{bmatrix} u_1^1 \\ u_2^1 \\ u_3^1 \end{bmatrix}=\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix},\quad \mathbf{v}^1=\mathbf{v}^c=\mathbf{v}^e=\begin{bmatrix} v_1^1 \\ v_2^1 \\ v_3^1 \end{bmatrix}=\begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix},$$

$$\mathbf{w}^1=\mathbf{w}^c=\mathbf{w}^e=\begin{bmatrix} w_1^1 \\ w_2^1 \\ w_3^1 \end{bmatrix}=\begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}. \tag{30}$$

*Forward Solution of Example 1.* The objective of forward kinematics is to find a final attitude from a set of given rotation angles. By substituting the three angles ($\alpha=90°$, $\beta=90°$, and $\gamma=90°$) into Eq. (26), the integrated rotation matrix can be found as

$$\mathbf{R}=\begin{vmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{vmatrix}. \tag{31}$$

Therefore, the attitude following the rotation can be calculated from Eq. (28) and presented as unit vectors of the body-fixed coordinate system

$$\mathbf{u}'=\mathbf{R}\mathbf{u}^0=\begin{vmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{vmatrix}\begin{vmatrix} 1 \\ 0 \\ 0 \end{vmatrix}=\begin{vmatrix} 0 \\ 0 \\ 1 \end{vmatrix},$$

$$\mathbf{v}'=\mathbf{R}\mathbf{v}^0=\begin{vmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{vmatrix}\begin{vmatrix} 0 \\ 1 \\ 0 \end{vmatrix}=\begin{vmatrix} -1 \\ 0 \\ 0 \end{vmatrix},$$

$$\mathbf{w}'=\mathbf{R}\mathbf{w}^0=\begin{vmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{vmatrix}\begin{vmatrix} 0 \\ 0 \\ 1 \end{vmatrix}=\begin{vmatrix} 0 \\ -1 \\ 0 \end{vmatrix}. \tag{32}$$

Comparing the results with graphic presentation in Fig. 5, and mathematical expressions in Eq. (30), it can be found

$$\mathbf{u}'=\mathbf{u}^1=\mathbf{u}^c=\mathbf{u}^e, \quad \mathbf{v}'=\mathbf{v}^1=\mathbf{v}^c=\mathbf{v}^e, \quad \mathbf{w}'=\mathbf{w}^1=\mathbf{w}^c=\mathbf{w}^e. \tag{33}$$

Therefore, the attitude of the humerus for the given rotation pattern is successfully determined. Since attitudes $(\mathbf{u}^c,\mathbf{v}^c,\mathbf{w}^c)$ and $(\mathbf{u}^e,\mathbf{v}^e,\mathbf{w}^e)$ were obtained from different sequences of rotation, the results demonstrate that the final attitude can be determined quantitatively in the spherical rotation coordinate system without considering the rotation sequences.

*Inverse Solution of Example 1.* The objective of inverse kinematics of human movement analysis is to find a set of rotation angles from recorded attitudes. In this example it is assumed that initial attitude given in Eq. (29) and final attitude given in Eq. (30) have been recorded. The purpose is to find three spherical rotation angles from the two known attitudes.

Since $\mathbf{a}_1=\mathbf{u}^1=\mathbf{u}^c=\mathbf{u}^e$, using Eqs. (3) and (4), long axis rotation angles $\alpha$ and $\beta$ can be calculated as

$$\alpha=\arccos\left(\frac{u_2^1}{\sqrt{(u_2^1)^2+(u_3^1)^2}}\right)=\arccos\left(\frac{0}{\sqrt{0+1^2}}\right)=90°,$$

$$\beta=\arccos(u_1^1)=\arccos(0)=90°. \tag{34}$$

Knowing the two angles $\alpha$ and $\beta$ the rotation matrix of the long axis rotation can be calculated from Eqs. (8) or (21) as

$$\mathbf{R}_l(90°,90°)=\begin{vmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{vmatrix}. \tag{35}$$

With the long axis rotation (abduction) unit vector $\mathbf{v}^0$ moves to $\mathbf{v}'$ that can be calculated using Eq. (10) as

$$\mathbf{v}'=\mathbf{R}_l(\alpha,\beta)\mathbf{v}^0=\mathbf{R}_l(90°,90°)\mathbf{v}^0$$
$$=\begin{vmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{vmatrix}\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}=\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}. \tag{36}$$

The axial rotation angle can then be determined using Eq. (11) as

$$\gamma=\arccos(\mathbf{v}'\cdot\mathbf{v}^1/|\mathbf{v}'||\mathbf{v}^1|)=\arccos(0)=90°. \tag{37}$$

The sign of the angles $\alpha$ and $\gamma$ can also be determined using Eqs. (5), (6), (12), and (13). Therefore, the three spherical rotation angles $\alpha=90°$, $\beta=90°$, and $\gamma=90°$ are fully determined.

*Example 2*

In this example, we investigate another pattern of humeral rotation involving 90° flexion and 90° external rotation, which is described as a 90° elevation at zero elevation plane and a 90° axial rotation in the spherical rotation coordinate system. The three angles are $\beta=90°$, $\alpha=0°$, and $\gamma=90°$. Graphical presentation of this pattern of rotation (Fig. 6) shows that the unique final attitude is

$$\mathbf{u}^2=\mathbf{u}^g=\mathbf{u}^i=\begin{bmatrix} u_1^2 \\ u_2^2 \\ u_3^2 \end{bmatrix}=\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{v}^2=\mathbf{v}^g=\mathbf{v}^i=\begin{bmatrix} v_1^2 \\ v_2^2 \\ v_3^2 \end{bmatrix}=\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix},$$

$$\mathbf{w}^2=\mathbf{w}^g=\mathbf{w}^i=\begin{bmatrix} w_1^2 \\ w_2^2 \\ w_3^2 \end{bmatrix}=\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}. \tag{38}$$

*Forward Solution of Example 2.* The three known rotation angles are $\beta=90°$, $\alpha=0°$, and $\gamma=90°$. Knowing the three rotation angles, the integrated rotation matrix can be found from Eq. (26) as

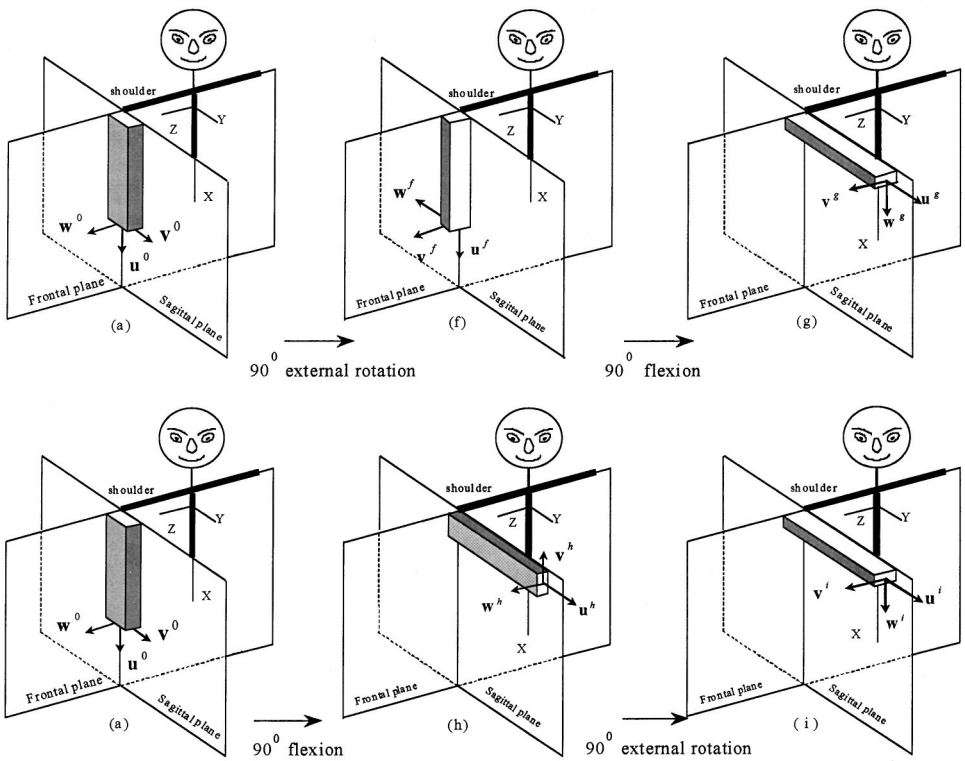


**FIGURE 6. A right humeral rotation involving 90° flexion and 90° external rotation can be performed in two sequences. In sequence 1, the first rotation is 90° external rotation from attitude ($\mathbf{u}^0,\mathbf{v}^0,\mathbf{w}^0$) in (a) to attitude ($\mathbf{u}^f,\mathbf{v}^f,\mathbf{w}^f$) in (f), then followed by 90° flexion to attitude ($\mathbf{u}^g,\mathbf{v}^g,\mathbf{w}^g$) in (g). In sequence 2, the first rotation is 90° flexion from attitude ($\mathbf{u}^0,\mathbf{v}^0,\mathbf{w}^0$) in (a) to attitude ($\mathbf{u}^h,\mathbf{v}^h,\mathbf{w}^h$) in (h), then followed by 90° external rotation to attitude ($\mathbf{u}^i,\mathbf{v}^i,\mathbf{w}^i$) in (i). As shown in the figure the two sequences are equivalent, since final attitude ($\mathbf{u}^g,\mathbf{v}^g,\mathbf{w}^g$) of sequence 1 equals to final attitude ($\mathbf{u}^i,\mathbf{v}^i,\mathbf{w}^i$) of sequence 2. *XYZ* is a reference coordinate system fixed on the trunk.**

$$\mathbf{R}=\begin{vmatrix}0&0&1\\1&0&0\\0&1&0\end{vmatrix}. \tag{39}$$

Therefore, the attitude following the rotation can be obtained using Eq. (28) as

$$\mathbf{u}''=\mathbf{R}\mathbf{u}^0=\begin{vmatrix}0&0&1\\1&0&0\\0&1&0\end{vmatrix}\begin{vmatrix}1\\0\\0\end{vmatrix}=\begin{vmatrix}0\\1\\0\end{vmatrix},$$

$$\mathbf{v}''=\mathbf{R}\mathbf{v}^0=\begin{vmatrix}0&0&1\\1&0&0\\0&1&0\end{vmatrix}\begin{vmatrix}0\\1\\0\end{vmatrix}=\begin{vmatrix}0\\0\\1\end{vmatrix},$$

$$\mathbf{w}''=\mathbf{R}\mathbf{w}^0=\begin{vmatrix}0&0&1\\1&0&0\\0&1&0\end{vmatrix}\begin{vmatrix}0\\0\\1\end{vmatrix}=\begin{vmatrix}1\\0\\0\end{vmatrix}. \tag{40}$$

The solutions are the same as known results in Eq. (38).

*Inverse Solution of Example 2.* It is assumed that the initial and final attitudes in Eqs. (29) and (38) have been recorded. Since $\mathbf{a}_1=\mathbf{u}^2=\mathbf{u}^g=\mathbf{u}^i$, the long axis rotation angles $\alpha$ and $\beta$ can be calculated using Eqs. (3) and (4) as

$$\alpha=\arccos\left(\frac{u_2^2}{\sqrt{(u_2^2)^2+(u_3^2)^2}}\right)=\arccos\left(\frac{1}{\sqrt{1+0}}\right)=0°,$$

$$\beta=\arccos(u_1^2)=\arccos(0)=90°. \tag{41}$$

Using Eq. (8) or Eq. (21) the rotation matrix for the long axis rotation is calculated as

$$\mathbf{R}_l(0°,90°)=\begin{vmatrix}0&-1&0\\1&0&0\\0&0&1\end{vmatrix}. \tag{42}$$

Using Eq. (10) the vector $\mathbf{v}'$ is calculated as

$$\mathbf{v}' = \mathbf{R}_l(0°,90°)\mathbf{v}^0 = \begin{vmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{vmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}. \quad (43)$$

The axial rotation angle can be determined using Eq. (11) as

$$\gamma = \arccos(\mathbf{v}' \cdot \mathbf{v}^2 / |\mathbf{v}'||\mathbf{v}^2|) = \arccos(0) = 90°. \quad (44)$$

These results are the same as known results.

## DISCUSSION

Based on a two-step rotation method, a spherical rotation coordinate system is developed in this paper for describing three-dimensional joint rotations. Two of the three spherical rotation angles—longitude and latitude—are the same as in the classical spherical coordinate system and they are used to define long axis rotations in the reference coordinate system. The spherical rotation coordinate system is different from the classical spherical coordinate system in that a radial rotation angle is used to define an axial rotation angle instead of a radial displacement.

The spherical rotation coordinate system inherited advantages of previous techniques. First, the two-step rotation used a concept of sequenced rotation in the Euler/Cardan angles, but reduces the three rotations in an Euler/Cardan sequence to two rotations in the new coordinate system. Therefore, there are only two possible sequences of rotations that are apparently sequence independent. Second, the determination of axial rotation in the two-step rotation method is similar to the instantaneous screw axis technique, which allows the axial rotation angle to be uniquely determined. Finally, geometrical positions of the long axis and an axis perpendicular to the long axis are considered in the determination of the three spherical rotation angles from recorded attitudes, which not only provides an advantage of easier visualization and interpretation of joint rotations but also avoids choosing proper axes in defining a joint coordinate system.

The major difference between the spherical rotation coordinate system and Euler/Cardan angles is that the long axis rotation is described in relation to the reference coordinate system and the axial rotation is described about a body-fixed axis (long axis), while all the three Euler/Cardan angles are either about the axes of a body-fixed frame or about the axes of a reference frame. The two long axis rotation angles—longitude and latitude—provide a concise depiction of the long axis position in a reference anatomical frame for clinical application. The definition of axial rotation angle is the same as the clinical description.

Using the spherical rotation coordinate system, a full range of three-dimensional joint rotations can be determined by the three spherical rotation angles at all positions except at the neutral attitude and the direct opposite of the neutral attitude (south and north poles). At these two attitudes, the angle $\alpha$ defining an elevation plane can have any value between $-180°$ and $180°$. Since elevation angle $\beta$ is zero at the neutral attitude and is $180°$ at the opposite of the neutral attitude, the angle $\alpha$ can be defined as *zero* at the two attitudes in most applications. However, care must be taken in assigning a proper value of the angle $\alpha$ at these two attitudes in some applications, such as forward simulation of joint rotations, to avoid discontinuity in computation. Special treatment may be required in applications that may have an elevation of over $180°$. In addition, the sensitivity in calculating long axis rotation angles near north and south poles may need to be considered in some applications.

Defining the long axis rotation angles $\alpha$ and $\beta$ with words is optional depending on the characteristics of joint and activity to be investigated. In the study of walking, it may be appropriate to use the procession or progression angle (i.e., medial/lateral rotation) for the longitude $\alpha$ at the hip and knee joints and to use the elevation angle (i.e., flexion/extension) for the latitude $\beta$. For the upper limb activities, as described earlier, the rotation of the long axis of the humerus can be illustrated by an elevation angle $\beta$ (latitude) and an angle of elevation plane $\alpha$ (longitude) or in more detail as medial/lateral rotation angles (Table 1) according to the humerus positions in the trunk frame.

This paper presented a mathematical proof that the long axis rotation and axial rotation are sequence independent. It is found that the unique integrated rotation matrix Eq. (26) can be obtained when reversing the order of multiplication of the long axis rotation matrix and the axial rotation matrix. However, it should be noted that rotation axis in the axial rotation matrices needs to be modified depending upon the position of the long axis. This is not contradictory to the general noncommutative property of two matrices multiplication. Also, the long axis rotation angle $\beta$ (latitude) can change at any given angle $\alpha$ (longitude). Given an initial attitude, except for the long axis at neutral and opposite of neutral attitudes (when $\beta = 0°$ and $\beta = 180°$), each of the three angles can change individually while the other two angles are fixed.

One of the important characteristics of the Euler/Cardan angles is the flexibility of positioning (in forward kinematics) a rigid body to a desired attitude in a reference coordinate system by combining a number of rotations about body-fixed axes in a specified sequence. An attitude ($\alpha$, $\beta$, $\gamma$) in the spherical rotation coordinate

system can also be obtained using a carefully composed sequence of Euler/Cardan angles. For example, the three spherical rotation angles can be obtained by a combination of four rotations: (1) rotate an angle $\alpha$ about the $x$ axis (long axis), (2) rotate an angle $\beta$ about the $z$ axis, (3) rotate an angle $-\alpha$ about the $x$ axis (at new position), and (4) rotate an angle $\gamma$ about the $x$ axis again, respectively. Three of the four rotations are about the same $x$ axis. If the third and fourth rotations were considered as one rotation $\varphi=\gamma-\alpha$, the four rotations would become the $x-z-x'$ sequence of Euler/Cardan rotation as reported in a previous study of glenohumeral rotations.[1]

The two simple examples given in this paper represent two different patterns of joint rotation that usually need to be described by two different sequences of Euler/Cardan rotations. The numerical results of the examples demonstrate effectiveness of the method of this paper in dealing with different patterns of joint rotations for both forward and inverse kinematic problems. Other patterns of joint rotations that are not limited within the sagittal or coronal planes can also be described in the spherical rotation coordinate system.

Definitions of the reference coordinate system (trunk system) and the body-fixed coordinate system (humeral system) in this paper are similar to that used by An *et al.*,[1] but are different to those proposed by Wu and Cavanagh[16] with the vertical axis of the trunk system directing downward. Since the elevation ($\beta$) of limb segments is usually measured from a neutral position when the limb segment is free hanging vertically, it is easier to observe the movement in the coordinate system defined in this paper with the long axis of the limb segment directed from the proximal joint to the distal joint and the vertical axis of the trunk frame directed downward.

In conclusion, the spherical rotation coordinate system provides a reliable and easily visualized description for three-dimensional joint rotations with three independent rotation angles that has a mathematical advantage over sequence dependent methods. The relevance of the three spherical rotation angles to anatomical description of joint rotations make them easy to be interpreted by clinicians for increasing understanding and assisting with the interpretation of human motion analysis data.

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